

# Exploring Customer Behavior Through Database Analysis

A SQL Case Study on Online Retailer Analytics, Visiting Patterns, and Loyalty  
Program Optimization.

PRESENTED BY DATA ANALYST TEAM



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# Introduction

Identifying key growth opportunities for online retailers by leveraging transactional data across five distinct relational datasets.



# THE ANALYTICAL INPUT

| Dataset Name | Core Attributes           | Analytical Purpose                         |
|--------------|---------------------------|--------------------------------------------|
| Sales        | User ID, Date, Product ID | Tracking order volume and timing.          |
| Product      | Product ID, Name, Price   | Revenue calculations and catalog mapping.  |
| Users        | User ID, Signup Date      | Calculating tenure and conversion metrics. |
| Gold Users   | User ID, Gold Signup Date | Analyzing loyalty program effectiveness.   |
| User Name    | User ID, Customer Name    | Personalizing insights and CRM linking.    |



# METHODOLOGIES & FRAMEWORK



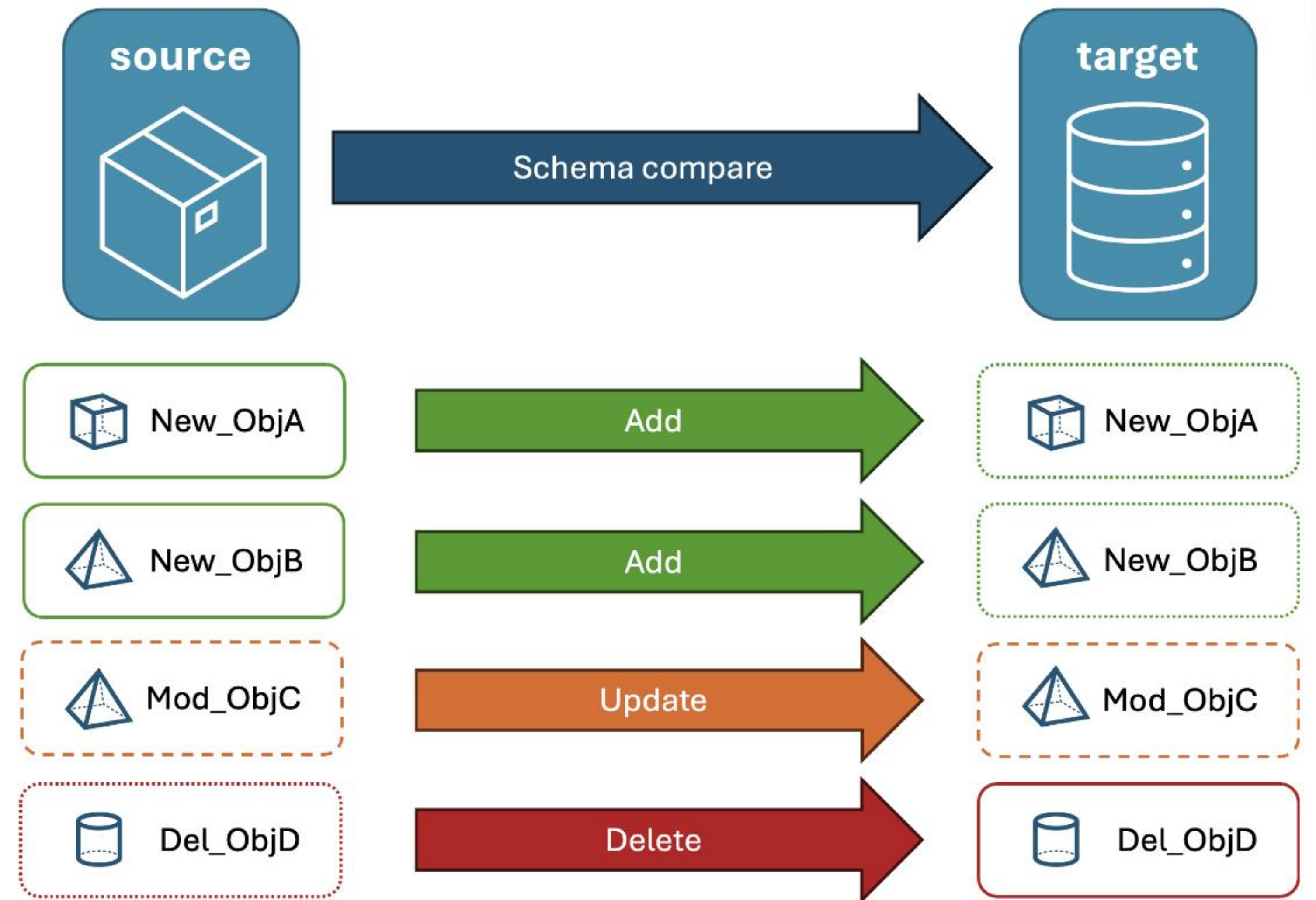
## SQL Aggregations

Utilizing SUM, COUNT, and AVG for high-level revenue and behavior summaries.



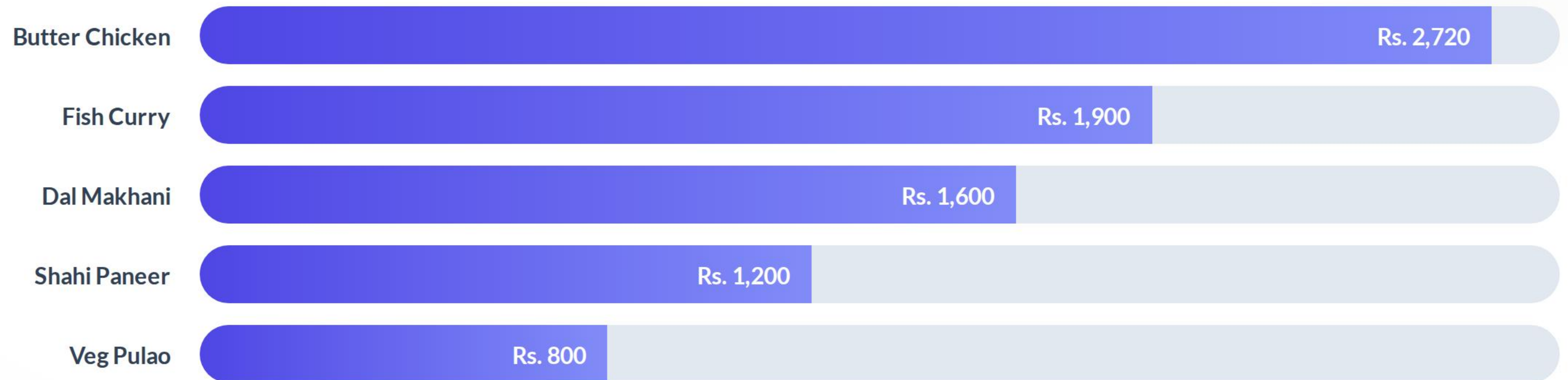
## Relational Joins

Complex INNER and LEFT JOINS to connect user profiles with transactional logs.





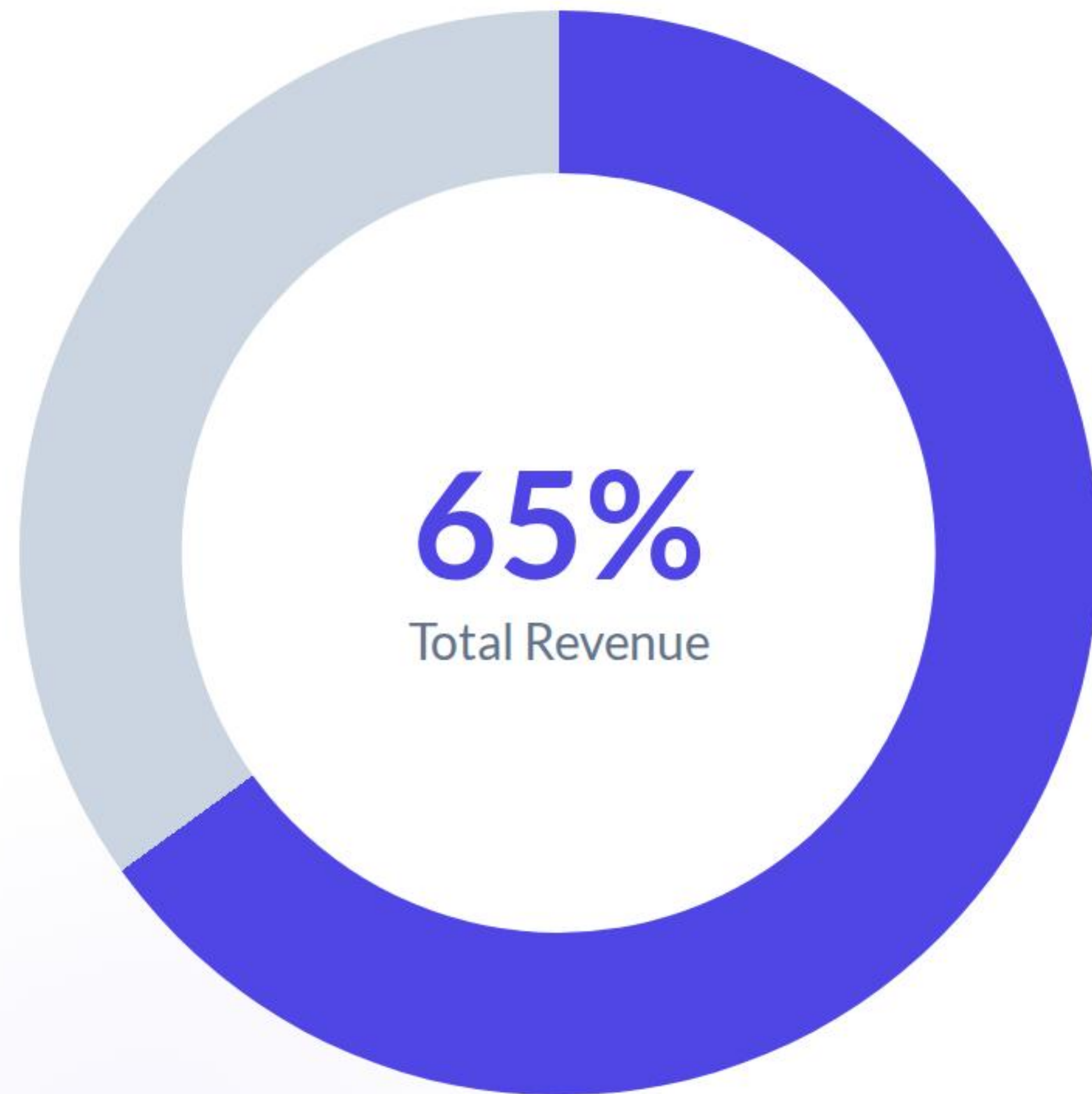
# KEY FINDING 1: REVENUE LEADERS



*Butter Chicken remains the primary revenue driver, generating significantly higher gross sales than any other menu item.*



## KEY FINDING 2: MEMBERSHIP VALUE



### Revenue Contribution

- Gold Members: 65% contribution
- Standard Users: 35% contribution

Despite representing a smaller user base, Gold Members generate the majority of total revenue.



## KEY FINDING 3: VISIT BEHAVIOR

12.5

Avg. Visits per Gold Member

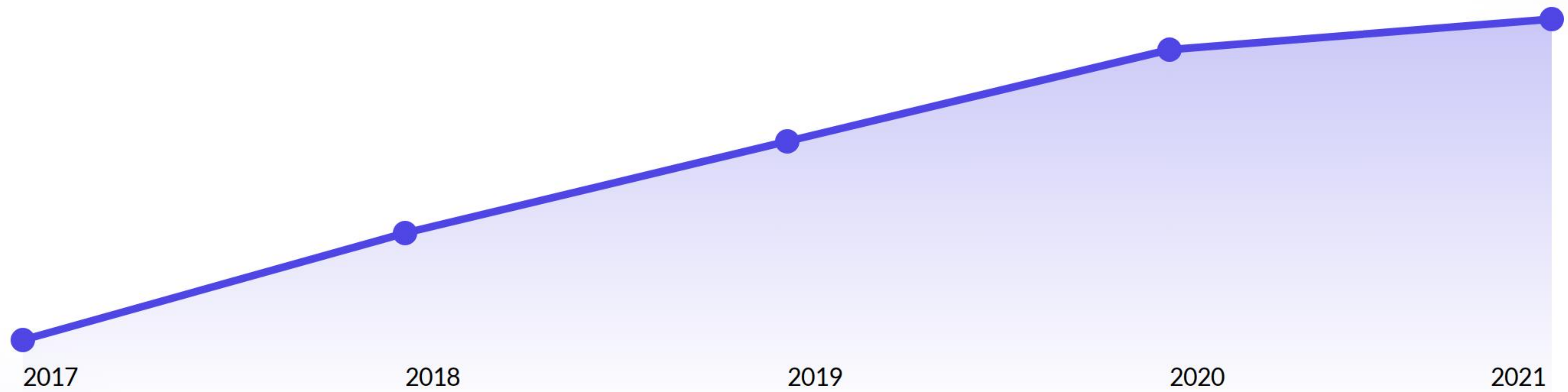
4.2

Avg. Visits per Standard User

Gold membership increases engagement frequency by nearly 3x compared to standard user signups.



## KEY FINDING 4: ANNUAL GROWTH



Steady YoY revenue growth indicates successful platform expansion and loyal customer base maturation.



## KEY FINDING 5: PRODUCT INDEX

- ★ **Most Purchased Item:** Butter Chicken holds the title for the highest volume across all segments.
- 👑 **Gold User Preference:** Premium members prioritize expensive curries and signature dishes.
- 🕒 **First Order Bias:** 40% of users start their journey with Dal Makhani or Dal Tadka.
- ↑ **Basket Growth:** Average order value (AOV) increases by 22% post-Gold signup.



## KEY FINDING 6: USER JOURNEYS



### Initial Signup

Generic browsing and small value first orders.



### Standard Use

Low frequency (4 visits/year) and price sensitivity.



### Gold Upgrade

Shift in behavior; higher frequency and spend.



### Loyal Advocate

Consistent high-value monthly active usage.



# STRATEGIC CONCLUSIONS

**Loyalty is Key:** Gold members are the backbone of the platform's financial stability.

**Menu Optimization:** Promote top-tier items (Butter Chicken) as gateways to membership.

**Data Transparency:** SQL-based automated reporting is essential for real-time decision making.

**Growth Potential:** Targeted marketing can convert standard users into the high-performing Gold tier.

PDF

## V – Conclusion (Short Version)

### Conclusion

This project titled "Information Technology & Its Application in Business" gave me a clear understanding of how Microsoft Word, Excel, and PowerPoint are used in day-to-day business activities. These tools are essential in the modern business world for handling documents, analyzing data, and giving presentations.

### What I Learned

#### MS Word

It is useful for creating business documents like letters, reports, and proposals. Formatting tools help make documents look professional and well-structured.

#### MS Excel

It is powerful for working with numbers, financial data, and reports. With functions, formulas, and charts, Excel helps in data analysis, accounting, and decision-making.

#### MS PowerPoint

It helps in presenting ideas in a clear and attractive way. With features like animations, images, and charts, presentations become more engaging and easy to understand.

### Key Observations

- All businesses use IT tools daily, whether small or large.
- These tools save time and reduce errors compared to manual work.
- Basic knowledge of Word, Excel, and PowerPoint is a must-have skill for any commerce student or employee.





# Questions & Answers

Thank you for your time. Feel free to ask any technical or strategic questions regarding the SQL Case Study.

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Data Analysis Project Portfolio



# IMAGE SOURCES



<https://learn.microsoft.com/en-us/sql/tools/sql-database-projects/concepts/media/schema-comparison/schema-compare-concept.png?view=sql-server-ver17>

Source: [learn.microsoft.com](https://learn.microsoft.com)



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